



**James' Super Cool  
IS 201 Cheat Sheet**

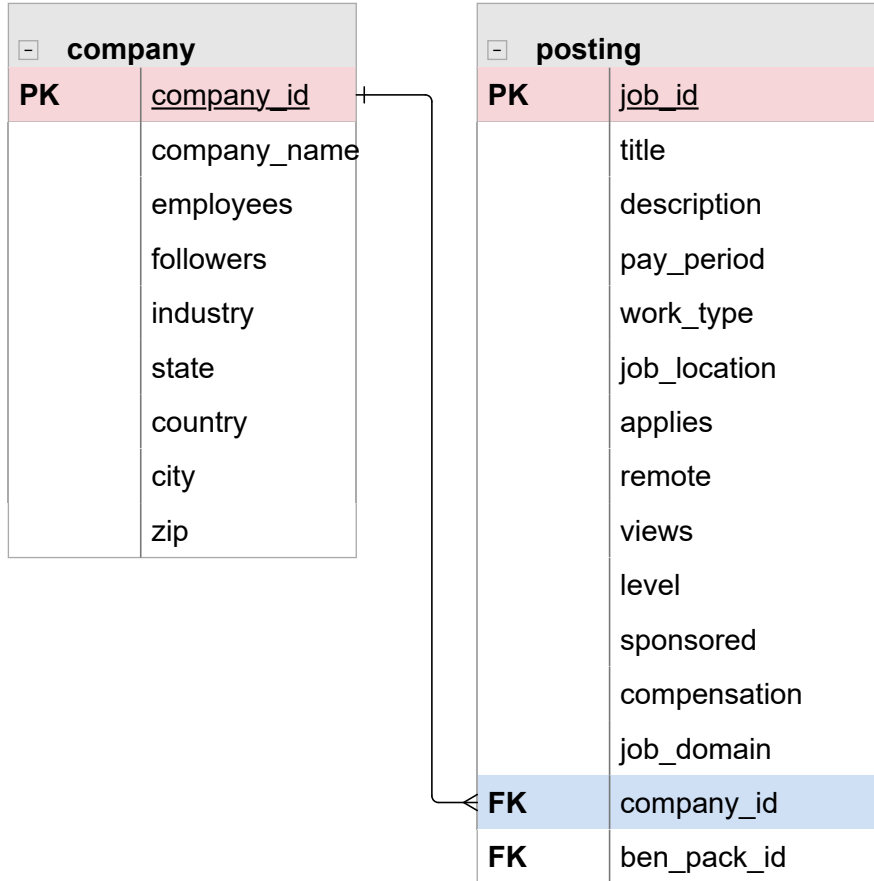
**You**



# Index

| Slide | Topic                              |
|-------|------------------------------------|
| 2     | <a href="#"><u>ERDs</u></a>        |
| 6     | <a href="#"><u>SQL</u></a>         |
| 8     | <a href="#"><u>Flow Charts</u></a> |
| 9     | <a href="#"><u>VBA</u></a>         |
| 15    | <a href="#"><u>Statistics</u></a>  |
| 16    | <a href="#"><u>Tableau</u></a>     |
| 17    | <a href="#"><u>Solver</u></a>      |
| 18    | <a href="#"><u>HTML</u></a>        |
| 27    | <a href="#"><u>CSS</u></a>         |

# ERD: Primary / Foreign Keys

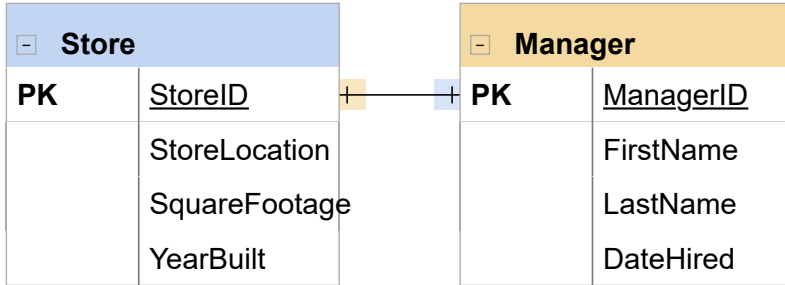


A **primary key** is a unique identifier (think social security number)

A **foreign key** is another table's primary key, used to link two tables together

Entity  
Relationship  
Diagram

# ERD: Cardinality

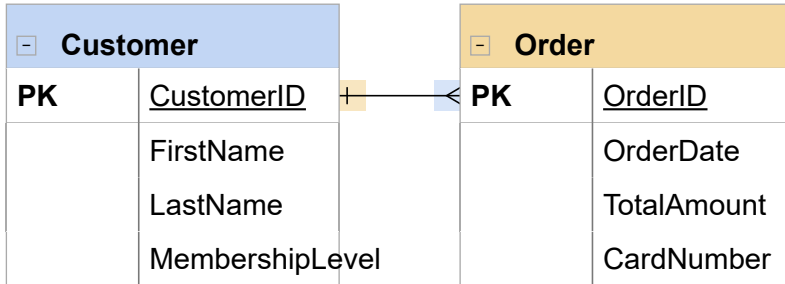


Read left-to-right AND right-to-left

## One to One (1:1)

A store has one manager

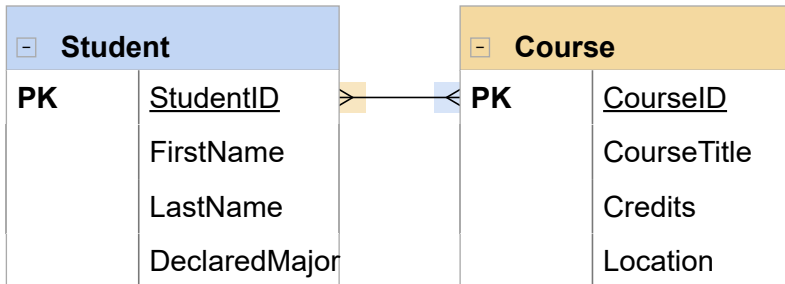
A manager works at one store



## One to Many (1:N)

A customer can have multiple orders

An order belongs to one customer



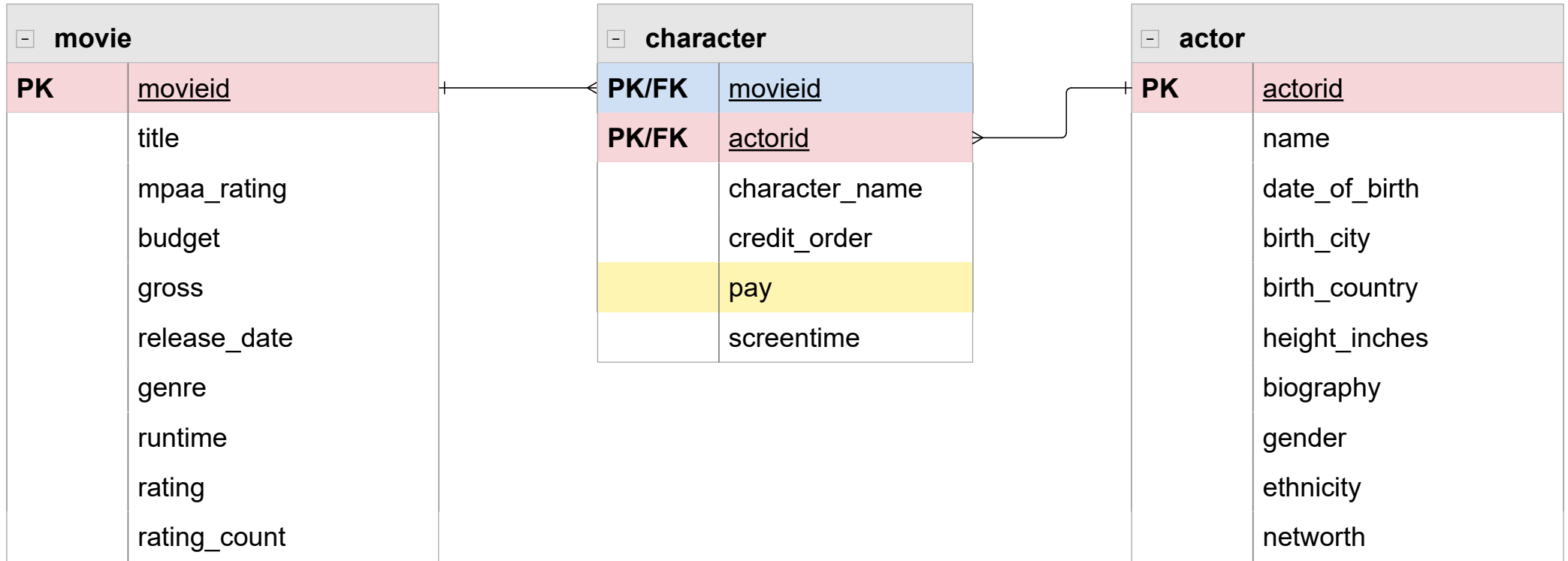
## Many to Many (M:N)

A student can enroll in many courses

A course can have lots of students

*Note! M:N cardinality requires a composite table (next slide)*

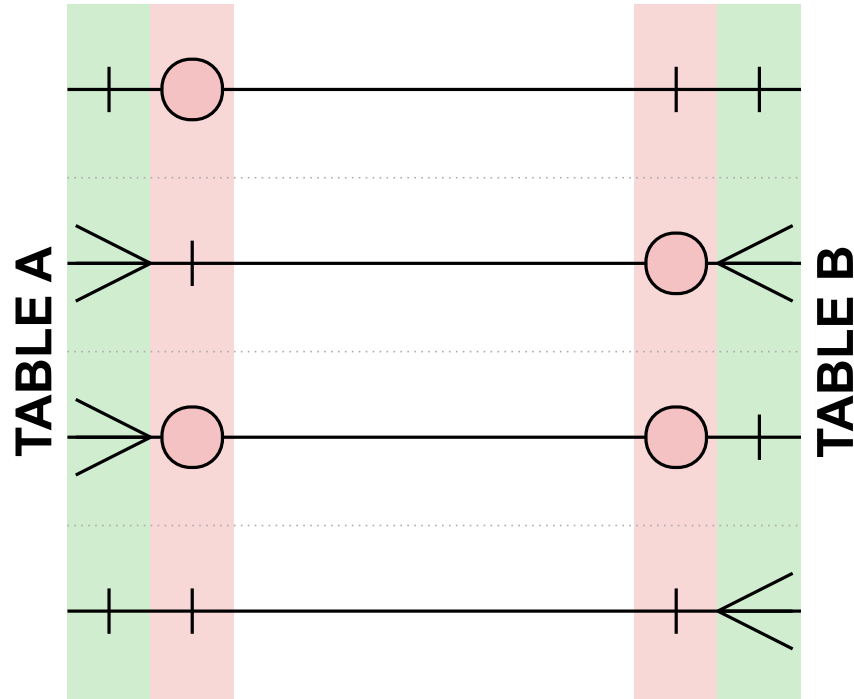
# ERD: Composite Table



To know how much money Tom Hanks got **paid** to play Woody in *Toy Story*, you need to know both the **movie** and the **actor**. His pay is probably different than when he was Woody in *Toy Story 2*, or when he was Forrest Gump in *Forrest Gump*

# ERD: Minimum / Maximum Cardinality

The inner marks are minimum (0 or 1)  
and the outer marks are maximum (1 or many)



A could have 1 B or 1 B

B could have 0 A or 1 A

A could have 0 B or many B

B could have 1 A or many A

A could have 0 B or 1 B

B could have 0 A or many A

A could have 1 B or many B

B could have 1 A or 1 A

*Think: a student could have 0 cars, but they could also have multiple.*

go to [gaskination.com/sql](https://gaskination.com/sql)

# SQL

```
1 SELECT      -- attributes -- SELECT * for all, SELECT DISTINCT for unique results
2 FROM        -- tableA
3 JOIN        -- tableB ON tableA.attribute = tableB.attribute (order doesn't matter)
4 WHERE       -- attribute filters [=, !=, <>, IS, LIKE '%__%', IN ("__","__","__")]
5     AND/OR  -- you only say WHERE once, but can have many filters
6 GROUP BY    -- all non-aggregated attributes when aggregating
7 HAVING      -- group-based filters (not used in the SQL project)
8 ORDER BY    -- attributes in ASC (default) / DESC order
9 LIMIT       -- to xx top results
10
11 -- aggregate(attribute) AS newName
12 count()
13 avg()
14 min()
15 max()
16 -- etc.
```

Structured  
Query  
Language

## SQL: Where Use Cases

ta\_name = "James"

→ Exact match

ta\_name != "James"  
ta\_name <> "James"

→ Not exact match  
*(these are equivalent)*

ta\_name **IS** NULL

→ NULL means blank data  
(can also do **IS NOT** NULL)

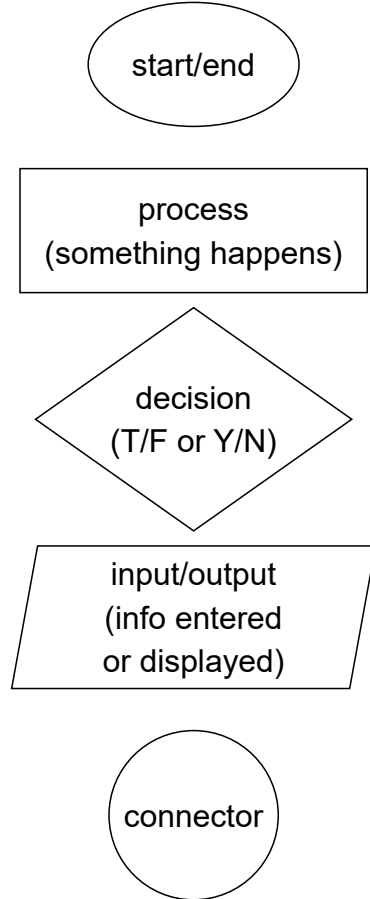
ta\_name **LIKE** '%ame%'

→ The text ("*ame*") is contained within the attribute  
(*\_ = one wildcard, % = any number of wildcards*)

ta\_name **IN** ("James", "Robert", "Frankie")

→ Exact match for any of these

# Flow Charts

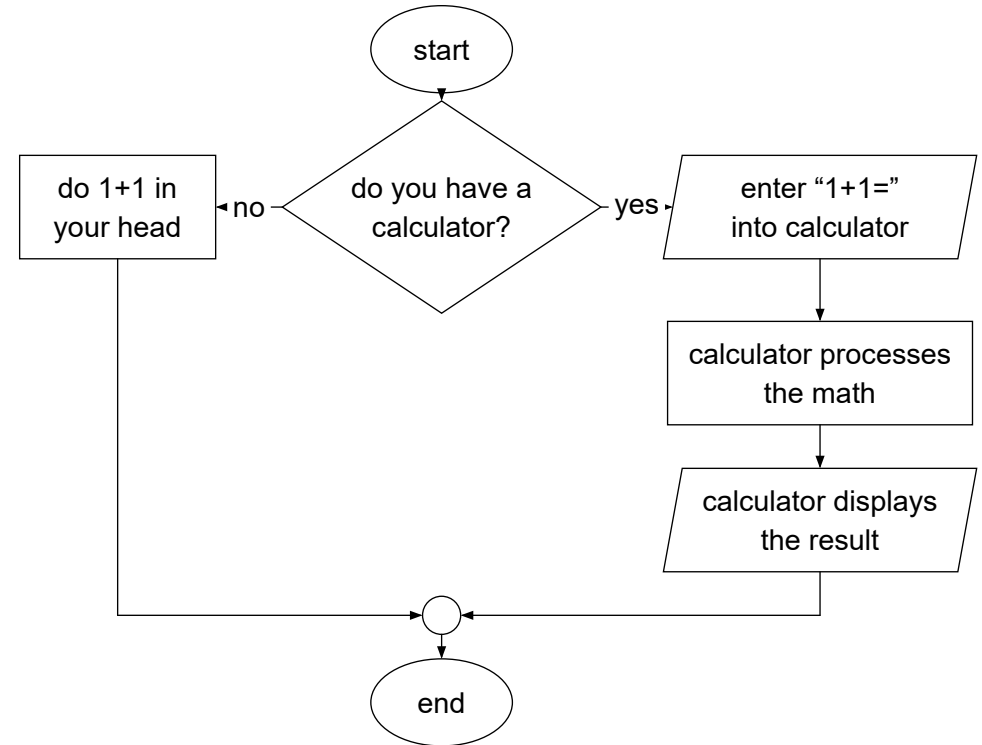


you can only have ONE start  
but you CAN have multiple  
ends

decisions are the ONLY  
thing that can have more  
than one arrow pointing OUT  
of them

connectors are the ONLY  
thing that can have more  
than one arrow pointing  
INTO them

## EXAMPLE: How to solve 1+1



# VBA: Basics

Option Explicit 'This makes it so that you can only use variables you've declared (VERY RECOMMENDED)

Sub thisIsMySubName()

    'BASICS!

    'Generally you will first refer to an object type (like a Sheet)

    'and then tell it what you want to do (like delete or add or copy, etc):

        ' Sheets and Worksheets are the same thing

        Sheets("MySheet").Delete

        Sheets.Add.name = "MySheet"

        Worksheets("MySheet").Activate

        ' Range and Cells are SIMILAR but different: Range("E7") = Cells(7, 5)

        Range("A1").Activate

        ActiveCell.Value = "Hello"

        Range("A1:D3").Copy

        Range("E5").PasteSpecial

        ' Columns and Rows are what they sound like

        Columns("B:D").Delete

Visual  
Basic for  
Applications

# VBA: Navigation and Misc.

'NAVIGATION!

```
'  
ActiveCell.End (xlDown)  
ActiveCell.End (xlToLeft)  
  
' Move (rows, columns) based on a cell or range  
ActiveCell.Offset(1, 2).Value = "NewSpot"  
    Range("A1").Offset(3, 4).Value = "NewSpot" 'The cell that now says "NewSpot" is E4
```

' MISCELLANEOUS!

```
' Bolding  
Range("A1").Font.Bold = True  
  
' Autofitting columns  
Columns.AutoFit 'all columns  
Columns("A:C").AutoFit 'specific columns  
  
' Currency format  
Range("A1").Style = "Currency"  
  
' Bottom border  
Range("A1").Borders(xlEdgeBottom).LineStyle = xlContinuous  
  
' Text splicing  
name = "James"  
first_initial = Left(name, 1)
```

# VBA: Variables

```
'DECLARE VARIABLES!
Dim i As Integer    'a whole number
Dim j As Double     'a number with a decimal
Dim k As String     'a line of text
Dim l As Boolean    'true or false
Dim m As Worksheet

i = 5
j = 1.2345
k = "Hello there! General Kenobi!"
l = True
Set m = Sheets("Sheet1") 'objects (sheets, books, ranges) use Set instead of =

'USER INTERFACE!

'Tell the user something
MsgBox "Oh hello there user, welcome to my spreadsheet"

'Ask for an input
Dim username As String
username = InputBox("What's your name?")

' COMBINING STRINGS

Dim myOutput As String
Dim name As String
name = "James"

' COMbine strings with ampersands (&)-- don't forget spaces
myOutput = "Hello. My name is " & name & " and this is Disney Channel!"
```

# VBA: Conditionals

'CONDITIONALS!

```
' If Statements
If i < 0 Then
    ' your code here
ElseIf i < 5 Then
    ' your code here
ElseIf i < 10 Then
    ' your code here
Else
    ' your code here
End If
```

```
'Case Statements
Select Case fruit
    Case "apple", "strawberry"
        Color = "red"
    Case "grape"
        Color = "purple"
    Case Else
        Color = "unknown"
End Select
```

```
'Another Case Statement
Select Case temperature
    Case Is >= 100
        feel = "too hot"
    Case 80 To 100
        feel = "pretty warm"
    Case Is < 80
        feel = "I'm probably staying out of the pool"
End Select
```

'LOOPS!

# VBA: Loops

```
'For Loop
For i = 0 To 5
    ' your code here
Next i
```

```
'For Each Loop
Dim x As Range
For Each x In Range("A2", Cells(lastrow, "A"))
    ' your code here
Next
```

```
'For Each Loop 2
Dim x As Range
Dim mtRange As Range
Set myRange = Range("A1:E5")
For Each x In myRange
    ' your code here
Next
```

```
'Do While Loop
Do While i = 5 'condition does NOT have to be numbers
    ' your code here
Loop
```

```
' Do Until Loop
Do Until i = 10 'condition does NOT have to be numbers
    'your code here
Loop
```

End Sub

# VBA: Functions

```
Function thisIsMyFunctionName(num As Integer, tf As Boolean) As String
```

```
' a function declaration is just like a set of dims  
' the INPUT is in the parentheses ()  
' the OUTPUT is outside of it  
' so this function KIND OF has this baked into it:
```

```
Dim num As Integer  
Dim tf As Boolean  
Dim thisIsMyFunctionName As String
```

```
' the INPUT then comes from your excel sheet  
' for example, in your sheet you cell might say:  
' =thisIsMyFunctionName(15, True)
```

```
' the OUTPUT comes from your function  
' set the function's name equal to something  
' in this case, our output has to be a string:
```

```
thisIsMyFunctionName = "That's a very nice hat you have there, partner!"
```

```
' generally your inputs will be used to calculate/decide your output
```

```
End Function
```

# Statistics

## TEST & USE CASE

### Correlation

numeric – numeric  
(Pearson  $r$  coefficient)

### Regression

numeric – multi. numeric

### ANOVA

numeric – mult. categorical

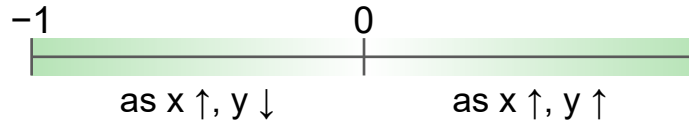
### T-test

numeric – 2 pair categorical

## OUTPUT

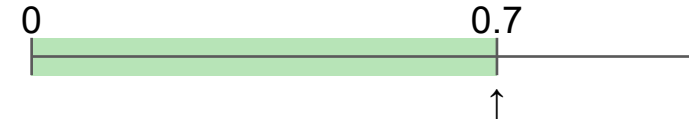
## HOW TO READ IT

→  $r$



*strength = distance from 0; sign = direction*

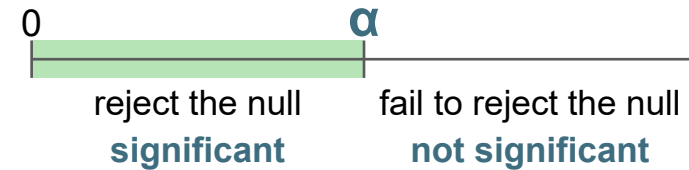
→  $R^2$



70% of the variance in  $y$   
is explained by the  $x$ 's

usually 0.05

→  $p$



*$p$  works like a percentage:  
0.05 = 5% to see this or more extreme*

## Null Hypothesis

$H_0$ : No  $x$ 's are  
significant

## Alt Hypothesis

$H_A$ : At least one  $x$  is  
significant

## Scientific Notation

$aE^b = a \times 10^b$

$aE3 = a \times 1000$  (big)

$aE-3 = a \times 0.001$   
(small)

## **Tableau**

- If the assignment says “default”, it means that you don’t have to change anything. It is the default setting inside Tableau
- Switching the x- and y-axis can reveal a clearer story (show an image of the button that does it)

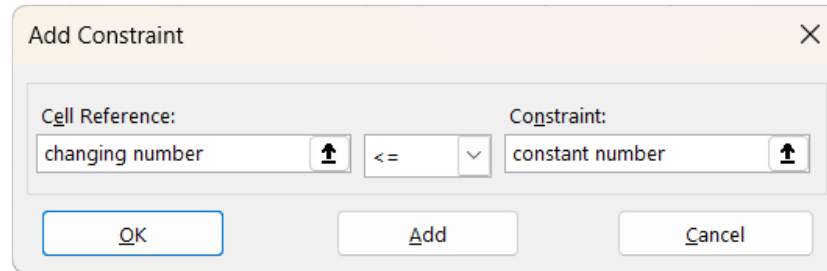
# Solver

## Two important Excel formulas:

=sum(A1:A5) →  $A1 + A2 + A3 + A4 + A5$

=sumproduct(A1:A5, B11:B15) →  $A1*B11 + A2*B12 + \dots + A5*B15$   
*the two ranges should be the same size and shape*

## Constraints:



Add Constraint

Cell Reference: changing number 

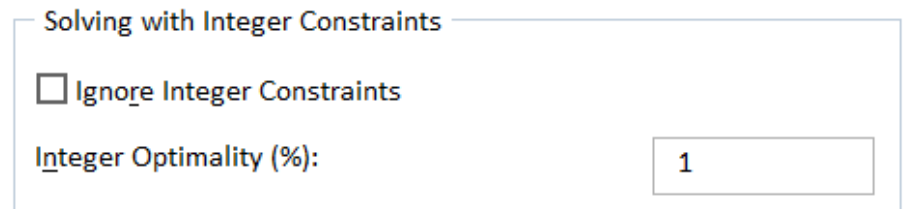
Constraint: constant number 

<=

OK Add Cancel

## If integer isn't working:

Solver > Options > uncheck this box:



Solving with Integer Constraints

☐ Ignore Integer Constraints

Integer Optimality (%): 1

# HTML: Setup

In an html file inside VS Code,  
type an exclamation point and enter to automatically generate:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>

</body>
</html>
```

Go here for free bootstrap themes:  
[startbootstrap.com/themes/portfolio-resume](https://startbootstrap.com/themes/portfolio-resume)

HyperText  
Markup  
Language

# HTML: Tags

```
<!-- This is a comment -->
<h1>Main heading</h1>
<h2>Section heading</h2>
<h3>Subsection</h3>

<p>This is a paragraph.</p>

<!-- Line Break -->
<br>
```

```
<!-- Unordered List (Bullets) -->
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>

<!-- Change <ul> to <ol>
for an ordered list (Numbers) -->
```

## Every tag follows this pattern:

|                     |                                |
|---------------------|--------------------------------|
| <b>&lt;tag&gt;</b>  | ← start tag                    |
| stuff               | ← what the tag applies to      |
| <b>&lt;/tag&gt;</b> | ← end tag: same name, plus a / |

*The start and end tags act like parentheses.*

(a few, like <br>, have no end tag)

# HTML: Anchors

```
<!-- Hyperlinks -->
```

```
<a href="https://www.google.com">Visit Google</a>
```

```
<!-- Folder References -->
```

```
<a href="otherpage.html">Visit Other Page</a>
```

```
<a href="subfolder/page.html">Visit Subfolder Page</a>
```

```
<a href="../anotherpage.html">Visit Another Page</a> ← ../ goes up one folder
```

```
<!-- On-page Anchors -->
```

```
<h2 id="contact">Contact Us</h2>
```

```
<a href="#contact">Jump to Contact Us</a> ← # + the id jumps to it on the same page
```

# HTML: Images

```
<!-- Images -->  
  
  

```

Notice: images do NOT have an end tag `</img>`  
*(neither do line breaks `</br>`)*

# HTML: Divisions

**Divs** don't inherently do anything: they're invisible boxes you put content into so that you can isolate it for positioning and CSS styling using classes.

HTML:

```
<div class="square-image">
  
</div>
```

CSS:

```
.square-image {
  width: 200px;
  height: 200px;
}
```

For more, see:

[CSS: Selectors/Properties/Values](#)

# HTML: Embedding

*Sometimes doesn't work until you go live*

## YouTube

1. Click **Share**
2. Click **Embed**
3. Click **Copy**
4. Paste into your code

## Tableau

1. Click the share icon
2. Click **Copy Embed Code**
3. Paste into your code

## HTML: Folder Structure & Formatting

**Use folders!** Keep `css/`, `html/`, `images/`, and `js/` separate, with `index.html` at the project root.

To quickly format everything correctly (tabs, long lines, etc.) in VS Code:

Right click → Format Document

## HTML: Uploading to GitHub

1. Click **New** to create a repository
2. Click **Create repository**
3. On the repo page: **Add file** → **Upload files**
4. Click **Commit changes**

**Upload the CONTENTS of your folder, not the folder itself**

## Going Live

1. Go to **Settings**
2. Click **Pages** in the sidebar
3. Under Branch, select `main` (instead of `None`) and **Save**

After a couple of minutes, refresh the page — there will be a link to your website at the top.

# CSS: Basics

```
1 <!-- Connect CSS to HTML, in your <head> -->
2 <link rel="stylesheet" href="styles.css" />
```



```
1 selector {
2     property: value;
3     property2: value2;
4 }
```



CSS is **cascading** because more recent styling supersedes previous styling — if you set the font color to blue, then later set it to red, it will be red.

Cascading  
Style  
Sheets

# CSS: Selectors / Properties / Values

```
1  /* Built-in selectors: the tags you use in your HTML (body, h1, p, a, img...) */
2  body {
3      background-color: #f0f0f0;
4      font-family: Arial, sans-serif;
5      margin: 0;
6      padding: 20px;
7  }
8  /* Class (custom) selectors: create your own with a period + name */
9  .highlight {
10     background-color: yellow;
11     font-weight: bold;
12 }
```

```
1 <p class="highlight">This text will be highlighted.</p>
```



*There are A LOT of selectors and properties — Google and AI are your friend for finding exactly what you want!*

[jimna-h.github.io/james\\_super\\_cool\\_is\\_201\\_cheatsheet](https://jimna-h.github.io/james_super_cool_is_201_cheatsheet)



IS 201 TA Lab